Aradhya Dwivedi

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Education

CSJM University Kanpur, India
Bachelor of Technology – Computer Science and Engineering; CGPA: 7.4 *July 2024 – June 2028*

Experience

Research Intern – Computer Vision & Satellite Imagery Summer 2026
Indian Institute of Technology (IIT) BHU *Varanasi, India*

- Built a **domain adaptation pipeline** that lets a **satellite image segmentation model** trained on one city (Dubai) work on a **completely different city** (Mumbai) without needing a full new labeled dataset.
- Trained **U-Net** with a **ResNet-34** encoder on Dubai aerial imagery, then ran a class-level **confusion matrix ablation** to pinpoint exactly why the model failed on Mumbai (structural blurring, spectral confusion between classes). Used that diagnosis to **design an encoder-frozen few-shot fine-tuning strategy** and used the approach by reading **IEEE** and **ICLR papers** on **domain adaptation**.
- Recovered mIoU from 0.07 (zero-shot, effectively broken) to 0.70, beating the 0.62 source-domain baseline and established that just 50 labeled images was the efficiency ceiling.

Projects

Pneumonia Detection System / *PyTorch, ResNet-18*

- Developed a deep learning-based medical image classification system using **ResNet-18** and **PyTorch** to detect pneumonia from chest X-rays.
- Built cross-language integration between a Python ML backend and a Java Swing GUI using ProcessBuilder, with crash-proofing mechanisms to prevent RAM leaks during heavy computational workloads.
- Live Demo: [add HuggingFace Space link here]

Smart Student Help Desk – AI Campus Assistant / *Antigravity IDE* [-4pt] itemize

Rapid-prototyped a unified **AI-powered student help desk** for CSJM University, covering service discovery, document guidance, and request tracking — built end-to-end in 3 hours at **BOB Hacks '26** (CSJMU, IBM, and the Institution's Innovation Council).

Implemented an AI chatbot for FAQ automation and document guidance, a request submission and ticket-tracking system, and personalized notifications, fully matching the hackathon's problem statement.

Independently architected the application logic, data flow, and technology stack, using Google Antigravity IDE's AI-assisted workflow to accelerate implementation.

Automated Web Extraction Engine & Data Pipeline / *Scrapy, MySQL, Python*

- Engineered a scalable web scraping ecosystem using the **Scrapy** framework to extract and process structured data from e-commerce platforms.
- Developed custom spiders for complex site architectures and implemented Item Pipelines for real-time data cleaning, validation, and storage in **MySQL**/JSON formats.
- Integrated middleware for user-agent rotation and proxy management, bypassing anti-bot measures to ensure reliability at scale.
- Executed a full-cycle **ETL** process, producing a structured dataset of 1,000+ products with pricing, stock, and rating attributes.

Technical Skills

Programming Languages: Python, Java, LaTeX
ML / Computer Vision: PyTorch, segmentation-models-pytorch, ResNet, U-Net, OpenCV, NumPy
Geospatial: Rasterio, GDAL (basics), Google Earth Engine (basics)
Web & Data: Scrapy, MySQL, FastAPI (basics), React (basics)
Tools: Git/GitHub, Google Colab, Overleaf, Antigravity IDE